When Calibrated Ensemble Fusion Fails Under Distribution Shift: A Reliability-Aware Study of Multi-Class Network Intrusion Detection

Abstract

Network intrusion detection systems are often evaluated using headline accuracy, yet operational cyber-defence requires more than correct hard labels: defenders need calibrated confidence estimates, reliable behaviour under attack-class imbalance, and mechanisms for identifying uncertain traffic that should be escalated for analyst review. This paper studies reliability-aware ensemble fusion for multi-class intrusion detection on the official UNSW-NB15 train/test split, focusing on calibration, model disagreement, leakage-controlled evaluation, and train/test distribution shift. The proposed diagnostic pipeline combines out-of-fold probability generation, per-base calibration, reliability-weighted averaging, multinomial stacking over full probability vectors and disagreement descriptors, and a sample-level gate trained to approximate the Brier-optimal mixture between stacked and weighted-average predictions. Across five stratified folds on the official training partition, the gated system attains macro $F_1 = 0.659 \pm 0.018$ and improves over single-model and simple-fusion comparators in paired fold-level comparisons. However, locked-test evaluation reveals a more cautious conclusion: the learned gate is less robust to the UNSW-NB15 train/test shift than simpler fusion, with p_{final} macro-F_1 falling by 0.143 between cross-validation and the locked split versus only 0.072 for plain weighted averaging. On the locked test set, plain weighted averaging achieves higher ensemble macro F_1 (0.549 vs 0.517) and base Extra Trees achieves the strongest overall macro F_1 (0.553), Brier score (0.029), ECE (0.042), and ROC-AUC (0.950). A selective-prediction analysis on p_{final} shows that escalating the bottom 10% of confidence raises retained-sample accuracy from 0.769 to 0.819 and macro F_1 from 0.517 to 0.565, illustrating how disagreement-aware confidence can support analyst triage. These results show that reliability-aware IDS fusion must be evaluated under locked-test distribution shift: a learned gate can improve cross-validated behaviour while degrading deployment-side macro F_1 relative to simpler fusion. The intended contribution is therefore a controlled reliability and failure-mode study of calibrated probability fusion, disagreement-aware stacking, and selective prediction for intrusion detection, not a claim that learned gating is the best deployment classifier on UNSW-NB15.

Keywords: Intrusion detection, UNSW-NB15, Ensemble learning, Probability calibration, Disagreement-aware gating, Feature stability, Class imbalance

1 Introduction

Network intrusion detection systems (NIDS) on the UNSW-NB15 corpus [**?**] are most often evaluated through a single static train/test split with overall accuracy or macro F_1 as the headline metric, and the literature is now saturated with reports of 97–99% binary accuracy. Yet the reliability of the predicted class probabilities, the behaviour of stacked ensembles under base-model disagreement, and the leakage discipline of the evaluation protocol are rarely audited. Three weaknesses recur in recent UNSW-NB15 ensemble work: base-model probabilities are typically left uncalibrated even when downstream stackers consume them as features; meta-learners are routinely fed only the per-model maximum probability, discarding cross-class structure and inter-model disagreement signals that are most informative precisely when the ensemble is least confident; and fold discipline is inconsistent, with selectors and calibrators sometimes fitted on the full training set before cross-validation. This paper takes the opposite stance: rather than chase the leaderboard, we build a calibrated, disagreement-aware ensemble whose contribution lies in decision quality under class imbalance.

This work is positioned within computer virology and hacking-technique defence because the operational objects being classified are traffic categories associated with exploitation, fuzzing, denial-of-service activity, reconnaissance, shellcode delivery, backdoor behaviour, and worm-like propagation. In such settings, the defender is not merely interested in the most likely label; the defender needs to know whether the model confidence is trustworthy enough to suppress, prioritise, or escalate an alert. A high-confidence false-benign or mis-prioritised attack prediction can produce missed compromise, delayed response, or analyst fatigue. We therefore treat UNSW-NB15 as a controlled benchmark for studying whether IDS probability estimates, model disagreement, and learned fusion components remain reliable when hacking-related traffic distributions shift between training and deployment.

This paper makes five contributions. **(1)** We define a leakage-controlled reliability protocol for multi-class UNSW-NB15 that separates cross-validated design behaviour from locked-test deployment behaviour. **(2)** We implement a calibrated disagreement-aware fusion pipeline combining out-of-fold probability generation, per-base calibration, reliability-weighted averaging, full-probability stacking, and sample-level gating. **(3)** We derive and evaluate a Brier-minimising oracle mixing target $\beta^*(x)$ for training a lightweight gate between stacked and weighted-average prediction paths. **(4)** We show that this learned gate is not robust under the official UNSW-NB15 train/test shift: despite favourable cross-validated behaviour, it loses locked-test macro F_1 to plain weighted averaging and base Extra Trees. **(5)** We demonstrate that the same calibrated confidence and disagreement signals remain operationally useful for selective prediction and analyst escalation, improving retained-sample accuracy and macro F_1 when low-confidence traffic is withheld from automatic classification.

2 Threat Model and Defensive Setting

We consider a defensive network-monitoring setting in which an IDS receives flow-level metadata and assigns each flow to one of ten UNSW-NB15 categories: *Normal*, *Generic*, *Exploits*, *Fuzzers*, *DoS*, *Reconnaissance*, *Analysis*, *Backdoor*, *Shellcode*, or

Worms. The attacker may generate traffic associated with exploitation, reconnaissance, denial-of-service activity, fuzzing, shellcode delivery, backdoor behaviour, or worm-like propagation. The defender does not rely on payload inspection in this study; classification is performed from structured flow features only.

The operational goal is not only to maximise hard-label accuracy, but also to produce reliable confidence estimates that can support alert prioritisation. A high-confidence incorrect prediction is more dangerous than a low-confidence incorrect prediction because it may cause a malicious flow to be ignored or mis-prioritised. Model disagreement is therefore treated as a defensive signal: when independently trained learners disagree, the sample may require escalation, manual analyst review, or secondary inspection. This framing motivates the paper's focus on calibration, Brier score, ECE, probability fusion, and disagreement-aware gating.

3 Related Work

Intrusion detection on UNSW-NB15.

Stacking, bagging and gradient-boosted ensembles dominate the recent UNSW-NB15 literature [?, ?, ?]. Zoghi and Serpen [?] combine Balanced Bagging, XGBoost and Random Forest with a Hellinger-distance criterion and threshold tuning, achieving 97.80% binary accuracy but reporting no calibration analysis, while Kasongo and Sun [?] compare feature-selection methods on UNSW-NB15 with XGBoost without examining the calibrated probability quality of the selected models. Recent IJIT contributions on intrusion detection [?, ?, ?] similarly emphasise accuracy and macro F_1 without auditing reliability or train/test shift sensitivity.

Calibration and reliability in security ML.

Platt scaling [?] and isotonic regression [?] remain the standard binary calibrators, and Niculescu-Mizil and Caruana [?] showed that boosted and bagged tree ensembles are systematically miscalibrated and benefit from post-hoc correction. For multi-class problems, Kull et al. [?] proposed Dirichlet calibration; we adopt the simpler one-vs-rest isotonic/sigmoid decomposition because it integrates cleanly with our OOF-fitting protocol. ECE binning, Brier-score decomposition, and reliability diagrams remain the standard tools for auditing IDS confidence quality.

Ensemble disagreement and dynamic selection.

The methodological ancestor of our stacker path is Wolpert's stacked generalisation [?], while the dynamic ensemble selection literature, exemplified by META-DES [?], learns per-instance competence from meta-features; our gate is closer in spirit to Mixture-of-Experts gating [?] in that it produces a continuous mixing coefficient between two pre-aggregated paths rather than selecting from a pool of base learners. Selective classification under uncertainty has been used to reduce risk by abstaining on low-confidence inputs; we evaluate an analogous escalation policy on the locked test set in §??. Aggregating multiple feature selectors to improve robustness is well-established [?, ?], but our selector adds a calibration-aware penalty and an explicit pairwise-Jaccard stability weight.

IDS and hacking-technique defence in JCVHT.

Intrusion detection has a direct history within the Journal of Computer Virology and Hacking Techniques. Morin and Mé [?] explicitly analysed the relationship between intrusion detection and computer virology, arguing that the two areas share complementary goals and techniques. Tripp [?] studied high-speed network IDS string matching through a parallel finite-state-machine design. More recent JCVHT work has evaluated machine-learning and ensemble-learning IDS models, including quantum machine-learning intrusion detection [?], explainable-AI analysis for IDS predictions [?], feature-selection-based IDS learning models [?], and ensemble-learning IDS for IIoT edge computing [?]. Compared with these works, the present paper does not primarily claim higher detection accuracy. Instead, it asks whether calibrated IDS probabilities, model disagreement, and learned fusion components remain reliable when attack-class priors and flow-feature distributions shift between training and deployment.

Gap addressed by this work.

Most IDS papers do not jointly audit calibration, leakage discipline, disagreement behaviour, and locked-test shift sensitivity. This leaves an operational gap: a classifier may appear strong on headline metrics while producing poorly calibrated confidence estimates or unstable behaviour on disagreement-heavy samples. Our work addresses this gap by treating ensemble probabilities as operational reliability signals rather than only intermediate quantities for hard-label prediction, and by explicitly quantifying how each component degrades between cross-validation and the locked split.

4 Dataset and Preprocessing

We use the official UNSW-NB15 train/test partition published by the Australian Centre for Cyber Security [?]: 82 332 training samples and 175 341 test samples, with 42 feature columns (after removing the synthetic `id` column) and two label columns, `attack_cat` (10-class multi-class target) and `label` (binary). Class distributions are highly imbalanced; Table **??** summarises the per-class counts.

Beyond class-distribution shift, the official UNSW-NB15 split also exhibits feature-distribution shift. Table **??** reports the ten numeric features with the largest absolute standardised mean difference (SMD) between training and test partitions, together with their Population Stability Index (PSI) computed on 10 training-quantile bins. The shifts are modest (max $|\mathrm{SMD}| \approx 0.15$) but non-negligible: TCP-handshake timing features (`tcprtt`, `synack`, `ackdat`), TTL-related features (`dttl`), TCP window features (`swin`, `dwin`), and connection-pattern aggregates all show train/test mean differences large enough to alter learned probability calibration on the test set.

A frozen schema specification fixes the column order, dtypes, and the multi-class label vocabulary; the loader rejects any input that violates the schema, strips whitespace and BOMs introduced by upstream tooling, and guarantees that label columns are disjoint from the feature column set. Because a single shared transformer cannot satisfy the inductive biases of trees, linear models, and distance-based learners simultaneously, we implement three preprocessing contracts: a *tree* pipeline (median

Table 1 UNSW-NB15 official-split class distribution. The official split is not class-distribution stationary: Normal traffic is substantially less common in the locked test set (−13.0 percentage points), while Exploits, Fuzzers, DoS, and Reconnaissance increase. The shift column is computed as test percentage minus train percentage. Multi-class classification is the primary task in this paper.

Class	Train	Train %	Test	Test %	Shift (pp)
Normal	37 000	44.94	56 000	31.94	−13.00
Generic	18 871	22.92	40 000	22.81	−0.11
Exploits	11 132	13.52	33 393	19.04	+5.52
Fuzzers	6 062	7.36	18 184	10.37	+3.01
DoS	4 089	4.97	12 264	6.99	+2.02
Reconnaissance	3 496	4.25	10 491	5.98	+1.73
Analysis	677	0.82	2 000	1.14	+0.32
Backdoor	583	0.71	1 746	1.00	+0.29
Shellcode	378	0.46	1 133	0.65	+0.19
Worms	44	0.05	130	0.07	+0.02
Total	82 332	100	175 341	100	—

Table 2 Top-10 most-shifted numeric features between train and test partitions of the official UNSW-NB15 split, ranked by absolute standardised mean difference (SMD). PSI is computed on 10 training-quantile bins; *n.a.* indicates degenerate quantile binning (e.g. heavy-tailed integer features). All values are computed directly from the official train/test CSVs.

Feature	Train mean	Test mean	|SMD|	PSI
`tcprtt`	0.0559	0.0414	0.146	0.023
`dttl`	95.71	79.61	0.142	0.021
`synack`	0.0293	0.0210	0.140	0.021
`swin`	133.46	116.26	0.135	n.a.
`ackdat`	0.0267	0.0204	0.130	0.045
`ct_dst_src_ltm`	7.46	8.73	0.114	0.056
`dwin`	128.29	115.01	0.104	n.a.
`ct_dst_sport_ltm`	3.66	4.21	0.093	0.024
`stcpb`	1.08×10^9	9.69×10^8	0.084	0.011
`rate`	82 411	95 406	0.083	0.111

imputation, ordinal encoding with explicit unknown values, no scaling) for Random Forest, Extra Trees and XGBoost; a *linear* pipeline (median imputation, RobustScaler, one-hot encoding with rare-category grouping at `min_frequency`= 5) for the stacker and linear selectors; and a *KNN* pipeline identical to the linear contract, since ordinal-coded categoricals must not be fed to a distance-based model. All transformers are fitted strictly on the training fold, with column order frozen at fit time and validated at transform time. To eliminate formula drift, every engineered feature is also centralised in an audited registry: each entry is a `FeatureSpec` with name, formula, input columns, category, and a clipping policy of either $[0, 1]$ (for ratio-valued features) or a train-fold-fitted percentile range. Table **??** lists the seventeen engineered features.

Table 3 Audited engineered-feature registry. $\varepsilon = 10^{-9}$ is added to every denominator to avoid division by zero without shifting scale. Clipping bounds for percentile policies are fitted on the training fold of each outer split.

Name	Formula	Category	Clipping
`byte_ratio`	sbytes/(sbytes + dbytes + ε)	ratio	$[0, 1]$
`pkt_ratio`	spkts/(spkts + dpkts + ε)	ratio	$[0, 1]$
`load_ratio`	sload/(sload + dload + ε)	ratio	$[0, 1]$
`ttl_diff`	sttl − dttl	asymmetry	pct. 0.5–99.5
`inter_pkt_ratio`	sinpkt/(sinpkt + dinpkt + ε)	asymmetry	$[0, 1]$
`jit_ratio`	sjit/(sjit + djit + ε)	asymmetry	$[0, 1]$
`mean_pkt_src`	sbytes/(spkts + ε)	packet size	pct. 0–99.5
`mean_pkt_dst`	dbytes/(dpkts + ε)	packet size	pct. 0–99.5
`pkt_size_asym`	mean_pkt_src/(mean_pkt_src + mean_pkt_dst + ε)	packet size	$[0, 1]$
`bytes_per_pkt_total`	(sbytes + dbytes)/(spkts + dpkts + ε)	packet size	pct. 0–99.5
`loss_rate`	(sloss+dloss)/(spkts+dpkts+ε)	loss	$[0, 1]$
`src_loss_rate`	sloss/(spkts + ε)	loss	$[0, 1]$
`dst_loss_rate`	dloss/(dpkts + ε)	loss	$[0, 1]$
`tcp_setup_ratio`	synack/(tcprtt + ε)	TCP	$[0, 1]$
`win_ratio`	swin/(swin + dwin + ε)	TCP	$[0, 1]$
`rate_per_pkt`	rate/(spkts + dpkts + ε)	load	pct. 0–99.5
`load_per_pkt`	(sload+dload)/(spkts+dpkts+ε)	load	pct. 0–99.5

5 Stability-Weighted Consensus Feature Selection

Ranking features by a single selector on a single split makes the chosen subset hostage to selector idiosyncrasy and split variance, so we aggregate four heterogeneous selectors over an inner stratified K-fold and weight each selector's contribution by its inner-fold predictive utility and the stability of its rankings. The pool spans complementary inductive biases — ANOVA F-test (univariate parametric), mutual information (univariate non-parametric), Random Forest importance (multivariate tree-based, captures interactions), and SGD ElasticNet coefficient magnitude (multivariate sparse linear) — all operating on the tree-preprocessed feature matrix to ensure consistent column ordering. Because the SGD ElasticNet selector operates on a representation originally designed for tree-compatible feature ordering, its coefficient magnitudes should be interpreted only as part of the consensus ranking procedure rather than as direct semantic evidence about ordinal-encoded categorical values; a future implementation could compute the ElasticNet selector on the linear one-hot preprocessing contract and map selected transformed features back to their source feature groups. For each candidate $k \in \{8, 12, 16, 20, 24, 30, 36, 42, 50, 57, 59\}$ and each selector s, three statistics are computed over an inner stratified K-fold (here $K = 3$) on the outer-training partition: the pairwise Jaccard stability $J_{s,k}$ across the $\binom{K}{2}$ inner-fold top-k sets; the inner-fold mean macro F_1 from a small surrogate Random Forest fitted on the top-k subset; and the inner-fold mean ECE of the same surrogate's probability

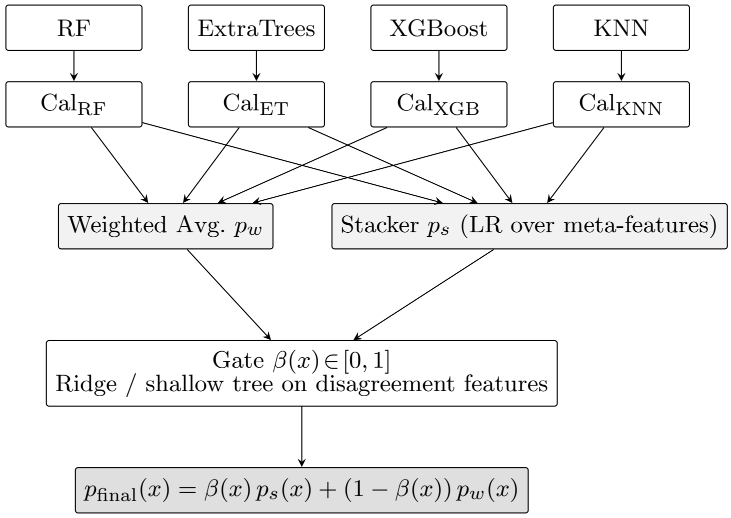

Fig. 1 Proposed architecture. Out-of-fold probability vectors flow from each base learner through a per-class isotonic/sigmoid calibrator, then into two parallel paths: a reliability-criterion weighted average and a multinomial-logistic stacker over disagreement meta-features. A sample-level gate trained to regress the closed-form optimal mixing coefficient $\beta^*(x)$ produces the final prediction.

output. The selector's score at k is

$$\Phi_{s,k} \;=\; \overline{F_1}_{\,s,k} \cdot \left(1 - \overline{\mathrm{ECE}}_{\,s,k}\right) \cdot J_{s,k}, \tag{1}$$

which jointly rewards predictive quality, calibration, and stability. The selector's chosen $k_s^* = \arg\max_k \Phi_{s,k}$ and weight $w_s = \max(\Phi_{s,k_s^*}, 0)$ define a convex contribution to feature-level consensus scores; the global k^* is the weighted average of k_s^* snapped to the candidate grid.

6 Calibrated Disagreement-Aware Ensemble

6.1 Architecture overview

Figure **??** sketches the architecture. Four base learners produce out-of-fold probability predictions; per-base calibrators correct miscalibration; the calibrated probabilities feed two parallel paths — a reliability-criterion weighted average and a multinomial-logistic stacker over disagreement meta-features — which are combined by a sample-level gate into the final prediction $p_{\mathrm{final}}(x)$.

6.2 Base learners

The base pool is Random Forest, Extra Trees, XGBoost, and KNN. Random Forest and Extra Trees are intentionally configured with orthogonal settings (RF: `max_features="sqrt"`, optimal split; ET: `max_features=1.0`, random split) to maximise inductive-bias diversity. XGBoost uses `multi:softprob` and the histogram tree method. KNN is retained for continuity with prior UNSW-NB15 ensemble work; the ablation study (§**??**) reports its honest contribution.

6.3 Out-of-fold probability collection

For each outer fold and each base learner, we run $K_{\mathrm{oof}} = 7$-fold stratified cross-validation on the outer-training partition. The preprocessor and the base estimator are refitted from scratch on each inner training fold; the held-out fold's probabilities are aligned to the global class index. Any class absent from a particular fold's training partition is assigned probability zero in that fold's predictions, and rows are renormalised post-hoc to lie on the simplex. The resulting $(n_{\mathrm{outer\_train}}, C)$ probability matrix is the only training substrate for calibration, the weighted averager, the stacker, and the gate.

6.4 Per-base calibration

Each base learner is wrapped in a one-vs-rest calibrator that fits one binary calibrator per class on the OOF probabilities. The choice of isotonic regression versus Platt scaling is made automatically per base, by comparing the multi-class Brier score on a validation split drawn from the OOF data; isotonic is the default when the OOF sample exceeds 1 000 rows. In our runs, isotonic was selected for every base in every outer fold. After calibration, each row is clipped to $[10^{-9}, 1]$ and renormalised to sum to unity.

6.5 Reliability-criterion weighted average

The weighted-average path computes

$$p_w(x) \;=\; \sum_{m=1}^{M} w_m\, \tilde{p}_m(x), \qquad w_m \;=\; \frac{\exp(\tilde{w}_m)}{\sum_{m'} \exp(\tilde{w}_{m'})}, \tag{2}$$

where the unnormalised score combines the OOF-evaluated macro F_1, log-loss, and ECE of base m:

$$\tilde{w}_m \;=\; \frac{F_1^{(m)} + \varepsilon}{\mathrm{LL}^{(m)} + \varepsilon} \cdot \max\!\big(1 - \mathrm{ECE}^{(m)},\, \varepsilon\big). \tag{3}$$

This rewards bases that are simultaneously accurate, low-loss, and well-calibrated. On the final training-partition refit, the resulting weights were $\{\mathrm{RF} : 0.242, \mathrm{ET} : 0.282, \mathrm{XGB} : 0.361, \mathrm{KNN} : 0.115\}$, with XGBoost dominating because of its near-zero ECE (0.003) post-calibration.

6.6 Disagreement-aware meta-features

The stacker and the gate consume the same meta-feature representation, built per sample from the calibrated probability matrices $\{\tilde{p}_m\}_{m=1}^{M}$. Four groups are concatenated:

1. *Per-model descriptors*: max probability, Shannon entropy, top-two margin (yields $3M$ scalars).
2. *Cross-model agreement*: per-class probability variance across models (yields C scalars), normalised plurality vote agreement, normalised vote entropy.
3. *Global confidence*: mean and variance of per-model max probabilities.
4. *Path-gap features* (gate only): absolute argmax-probability gap between p_s and p_w, agreement indicator on the predicted class, and per-class absolute path difference.

By default the meta-feature vector also includes the full calibrated probability vectors $\{\tilde{p}_m\}$ themselves; an ablation (§**??**) replaces them with scalar descriptors only.

6.7 Stacker path

The stacker is multinomial logistic regression with L_2 regularisation, class weighting balanced, and a small gradient-boosting fallback automatically selected when the LR's three-fold OOF Brier exceeds the GB's by more than 0.005. In every outer fold of our main experiment the LR was retained: the GB stacker had a slightly lower OOF Brier in absolute terms (≈ 0.013–0.014 versus ≈ 0.018 for LR), but never by more than the 0.005 improvement margin required to switch.

6.8 Disagreement-aware gate

The gate produces a per-sample mixing coefficient $\beta(x) \in [0, 1]$ such that

$$p_{\text{final}}(x) \;=\; \beta(x)\, p_s(x) + (1 - \beta(x))\, p_w(x). \tag{4}$$

We frame gate training as regression to a closed-form optimal target. For a labelled sample with one-hot encoding $y_{\text{oh}} \in \{0, 1\}^C$, weighted-average prediction p_w, and stacker prediction p_s, the gate target minimises the squared error

$$\mathcal{L}(\beta) \;=\; \left\| \, y_{\text{oh}} - (\beta\, p_s + (1 - \beta)\, p_w) \, \right\|_2^2. \tag{5}$$

Letting $d = p_s - p_w$, the candidate mixture rewrites as $p_w + \beta\, d$, so $\mathcal{L}(\beta) = \|(y_{\text{oh}} - p_w) - \beta\, d\|_2^2$. Differentiating and setting to zero,

$$\frac{d\mathcal{L}}{d\beta} \;=\; -2\, \langle y_{\text{oh}} - p_w,\, d \rangle + 2\, \beta\, \|d\|_2^2 \;=\; 0, \tag{6}$$

yields the unconstrained optimum $\beta_{\text{unclipped}} = \langle y_{\text{oh}} - p_w,\, p_s - p_w \rangle / (\|p_s - p_w\|_2^2 + \varepsilon)$. Clipping to the unit interval gives the deployed target

$$\beta^*(x) \;=\; \text{clip}_{[0,1]}\left(\frac{\langle\, y_{\text{oh}} - p_w,\; p_s - p_w \,\rangle}{\|\, p_s - p_w \,\|_2^2 + \varepsilon} \right), \tag{7}$$

which is well-defined whenever the two paths disagree. The gate is then a lightweight regressor (Ridge or shallow decision tree, max_depth ≤ 4, automatically chosen by inner CV MSE) that maps disagreement meta-features to β^*. Crucially, the gate consumes only meta-features — it never sees raw traffic features — which keeps it interpretable and prevents it from re-learning the base task. In our final training-partition run a depth-4 tree was selected over Ridge, and the empirical distribution of β^* was strongly bimodal (mean 0.481, std 0.492): the optimal mixture is most often near 0 or near 1, validating the design intuition that fixed mixing is suboptimal.

Algorithm 1 One outer fold of the proposed pipeline

Split outer_train / outer_valid; fit `FeatureEngineer` on outer_train only.
Run `ConsensusSelector` on outer_train (inner K-fold) $\Rightarrow$ chosen feature set.
Restrict outer_train and outer_valid to the selected features.
(Optional) RandomizedSearchCV per base learner on outer_train; persist best params.
for each base learner m **do**
  Collect OOF probabilities $\{p_m\}$ via K_{oof}-fold stratified CV on outer_train.
  Fit OvR calibrator C_m on $(p_m, y_{\text{outer\_train}})$; obtain calibrated $\tilde{p}_m$.
end for
Fit `WeightedAverager` on $\{\tilde{p}_m\}$ using Eq. **??**; produce p_w on outer_train.
Build meta-features $M(\tilde{p}_{1:M})$; obtain stacker OOF p_s via inner 3-fold on M.
Augment meta-features with path-gap; compute β^* from Eq. **??** and fit the gate.
Refit all base learners on full outer_train (no preprocessing leakage).
Evaluate $p_w, p_s, p_{\text{final}}$ and every base/calibrated path on outer_valid.

6.9 Final locked-test training procedure

After all design choices were fixed using the official training partition, the final locked-test model was trained only on that training partition. The feature engineer and consensus selector were refit on the training partition only. Base learners were refit using the selected features and the frozen preprocessing contracts. For the locked-test model, hyperparameters were selected using only the official training partition: we used the hyperparameter configuration with the best mean inner-CV macro F_1 across the outer-fold `RandomizedSearchCV` runs, breaking ties by lower log loss. Out-of-fold predictions on the training partition were then generated using the same seven-fold protocol to train the calibrators, weighted-average weights, stacker, and gate. The final base learners were refit on the full training partition, and the learned calibration and fusion components were applied once to the locked test set. The locked test labels were never used for feature selection, calibration, hyperparameter selection, model selection, stacker training, or gate training.

6.10 Hyperparameter search space and implementation details

Table **??** lists the per-model search space; `RandomizedSearchCV` samples 20 configurations from each. Implementation: Python 3.x with scikit-learn, XGBoost, NumPy, pandas, and SciPy; deterministic seed registry with global seed 42; all artefacts (model files, calibration reports, run manifest, fold-level metrics) are stored under artifacts/<run_id>/ and the figures under reports/<run_id>/. The full reproducibility manifest, including library versions used at training time, is recorded in the run-manifest YAML for each experiment.

Table 4 Per-model hyperparameter search spaces used by `RandomizedSearchCV` (n_iter = 20, three-fold inner CV, scoring `f1_macro`). The default values used as the starting point of the search space are documented in the codebase.

Model	Search space
Random Forest	`n_estimators` $\in \{100, 200, 300, 400, 500\}$; `max_depth` $\in \{\text{None}, 10, 15, 20, 30\}$; `min_samples_split` $\in \{2, 5, 10, 20\}$; `max_features` $\in \{\text{sqrt}, \text{log2}\}$; `class_weight`=`balanced`.
Extra Trees	`n_estimators` $\in \{100, 200, 300, 400, 500\}$; `max_depth` $\in \{\text{None}, 10, 15, 20, 30\}$; `min_samples_split` $\in \{2, 5, 10, 20\}$; `max_features` $\in \{0.3, 0.5, 0.75, 1.0\}$; `class_weight`=`balanced`.
XGBoost	`n_estimators` $\in \{200, 300, 400, 500\}$; `max_depth` $\in \{4, 5, 6, 8, 10\}$; `learning_rate` $\in \{0.01, 0.05, 0.1, 0.15, 0.2\}$; `subsample` $\in \{0.6, 0.7, 0.8, 0.9, 1.0\}$; `colsample_bytree` $\in \{0.6, 0.7, 0.8, 0.9, 1.0\}$; `min_child_weight` $\in \{1, 3, 5, 7\}$.
KNN	`n_neighbors` $\in \{5, 10, 15, 20, 25, 30\}$; `metric` $\in \{\text{euclidean}, \text{manhattan}\}$; `weights`=`distance`.

6.11 Class-imbalance handling

UNSW-NB15 is severely imbalanced (`Worms` has only 44 training samples). We handle imbalance with class weighting wherever the learner supports it: Random Forest, Extra Trees, the surrogate Random Forest used inside the consensus selector, the multinomial-logistic stacker, and the linear SGD ElasticNet selector all use `class_weight="balanced"`. XGBoost is trained without explicit class weighting because the tree booster's softmax objective combined with macro-averaged scoring during hyperparameter search is sufficient in our experiments; no per-class `scale_pos_weight` is applied in the multi-class objective. KNN is left with uniform sample weighting and distance-based neighbour weighting (`weights="distance"`), since a distance-based learner has no native class-balancing mechanism. We report macro F_1 rather than accuracy as the primary metric because accuracy is dominated by the four largest classes, which together account for 84.2% of the test set; minority classes such as `Analysis` and `Backdoor` therefore contribute almost nothing to accuracy and are best assessed through macro F_1, per-class F_1, and recall.

7 Experimental Protocol

7.1 Folds and hyperparameter search

We use five outer stratified folds on the official UNSW-NB15 training partition, three inner stratified folds for consensus selection, and seven-fold stratified OOF probability collection per base learner. The locked official test set is used exactly once, after all design decisions have been frozen, to produce the reported test metrics. Inside each outer fold we run `RandomizedSearchCV` (n_iter = 20, three-fold inner CV, scoring `f1_macro`) per base learner on the selected-feature matrix; the tuned parameters used for the final locked-test refit were RF $\{500, \text{depth}{=}30, \text{sqrt}\}$, ET $\{500, \text{depth}{=}20, \text{maxfeat}{=}0.3\}$, XGB $\{400, \text{depth}{=}5, \text{lr}{=}0.1, \text{colsample}{=}0.7\}$, and KNN $\{k{=}5, \text{Manhattan}\}$.

7.2 Metrics and ablations

The reported metrics are macro F_1 (primary), weighted F_1, accuracy, multi-class log-loss, multi-class Brier score, Expected Calibration Error (ECE), and macro one-vs-rest ROC-AUC. ECE is computed using fifteen equal-width bins over the maximum predicted class probability; for each non-empty bin, confidence is the mean maximum predicted probability and accuracy is the fraction of samples whose predicted class matches the true class, and bin gaps are weighted by the fraction of samples in the bin. The reported multi-class Brier score for C classes and n samples is

$$\text{Brier} \;=\; \frac{1}{C}\sum_{c=1}^{C}\frac{1}{n}\sum_{i=1}^{n}\big(p_{ic} - \mathbb{1}[y_i = c]\big)^2. \tag{8}$$

Nine ablation configurations isolate the contribution of each component: `no_calibration`, `no_gate` (fixed $\beta = 0.5$), `stacker_only` ($\beta = 1$), `weighted_avg_only` ($\beta = 0$), `simplified_meta_features` (scalar descriptors only), `no_engineered_features`, `no_feature_selection`, `no_stability_weighting`, and `single_selector` (mutual information only). A deterministic seed registry maps every stochastic component to a sub-seed derived from a single global seed (default 42), and the full suite runs end-to-end from one script.

8 Results

8.1 Selector behaviour

The four heterogeneous selectors converged on near-identical weights under the calibration-aware criterion of Eq. **??** and selected the same feature-count $k^* = 50$ in every outer fold (Table **??**). Among the top eight ranked features were five engineered ones — `load_per_pkt`, `rate_per_pkt`, `win_ratio`, `tcp_setup_ratio`, `bytes_per_pkt_total` — alongside the categorical `state`, `service`, and `proto`.

Table 5 Empirical selector behaviour from the final locked-test refit on the full training partition. The four selectors converge on near-identical weights, evidence of broad agreement on which features carry predictive signal under the calibration-aware penalty.

Selector	Weight w_s	Chosen k_s^*
ANOVA F-test	0.460	50
Mutual information	0.479	50
Random Forest importance	0.474	50
SGD ElasticNet	0.473	50
Consensus k^*	—	50

8.2 Cross-validated multi-class results

The proposed calibrated gated system attains macro $F_1 = 0.659 \pm 0.018$. It improves over the simple fusion baselines and every single-model baseline in the paired fold-level comparisons, but several diagnostic ablations obtain slightly higher cross-validated macro F_1. In particular, the no-calibration, no-stability-weighting, and no-engineered-feature variants obtain macro F_1 values of 0.668, 0.665, and 0.663, respectively. We therefore interpret the full system not as the maximal hard-label macro-F_1 configuration, but as the most complete reliability-oriented configuration balancing macro F_1, calibration, Brier score, and disagreement-aware behaviour. Removing the gate (fixed $\beta = 0.5$) degrades macro F_1 by 0.017; replacing the meta-features with scalar descriptors costs 0.045; restricting the gate to the stacker path alone ($\beta = 1$) costs 0.055. Table **??** reports five-fold cross-validation means and standard deviations on the training partition for the proposed system and every ablation.

Table 6 Five-fold cross-validated multi-class results on the UNSW-NB15 training partition. Values in parentheses are standard deviations across folds. Bold marks the proposed system p_{final} for orientation only; three diagnostic ablations (no_calibration, no_stability_weighting, no_engineered_features) obtain slightly higher CV macro F_1, which we discuss in §**??**.

Configuration	mac. F_1	wgt. F_1	acc.	log loss	ECE	ROC-AUC
p_{final} **(proposed)**	**0.659 (.018)**	0.888 (.001)	0.884 (.004)	0.309 (.004)	0.023 (.003)	0.982 (.004)
no_calibration	0.668 (.014)	0.892 (.003)	0.884 (.005)	0.310 (.016)	0.019 (.001)	0.984 (.003)
no_stability_weighting	0.665 (.011)	0.888 (.002)	0.884 (.001)	0.309 (.004)	0.024 (.002)	0.981 (.007)
no_engineered_features	0.663 (.012)	0.889 (.002)	0.886 (.001)	0.308 (.004)	0.024 (.004)	0.982 (.002)
single_selector (MI)	0.654 (.009)	0.889 (.001)	0.882 (.004)	0.311 (.005)	0.025 (.003)	0.979 (.007)
no_feature_selection	0.651 (.014)	0.890 (.002)	0.881 (.002)	0.311 (.003)	0.026 (.002)	0.981 (.006)
no_gate (β=0.5)	0.642 (.007)	0.889 (.002)	0.879 (.002)	0.319 (.005)	0.029 (.001)	0.981 (.006)
weighted_avg (β=0)	0.621 (.007)	0.884 (.003)	0.886 (.002)	0.305 (.005)	0.034 (.001)	0.981 (.001)
simplified_meta	0.614 (.028)	0.886 (.003)	0.888 (.004)	0.304 (.014)	0.030 (.003)	0.983 (.001)
stacker_only (β=1)	0.604 (.008)	0.880 (.002)	0.866 (.001)	0.405 (.005)	0.023 (.003)	0.972 (.013)

Statistical tests.

Because only five outer folds are available, the paired tests in Table **??** should be interpreted as descriptive evidence of consistent fold-level direction rather than definitive inferential proof. The Wilcoxon signed-rank p-value reaches 0.0312 in every comparison because this is the smallest attainable two-sided p-value with $n = 5$ paired observations when all fold differences have the same sign. Subject to this caveat, p_{final} improves on every single-model and simple-fusion comparator on macro F_1 at $p < 0.05$, and all ten paired t-test p-values remain below their per-rank Holm–Bonferroni thresholds at $\alpha = 0.05$. We use these tests to support the cross-validated diagnostic comparison, while treating the locked-test paired bootstrap analysis (§**??**) as the stronger evidence for deployment-side conclusions. We additionally compared p_{final} against the three ablations that achieved slightly higher CV macro F_1: p_{final} does *not* significantly differ from the no_calibration variant (paired t $p = 0.073$, Wilcoxon $p = 0.062$), the no_stability_weighting variant ($p = 0.298$, $p = 0.438$), or the no_engineered_features variant ($p = 0.356, p = 0.312$), so the small CV-macro-F_1 advantages of those ablations are within fold-level noise.

Table 7 Paired statistical tests on macro F_1 across the five outer folds. p_{final} **wins every comparison.** Wilcoxon $p = 0.0312$ is the saturation lower bound for $n = 5$ paired observations.

Baseline	mean diff	t-statistic	t p-value	Wilcoxon p
p_{weighted}	0.0384	7.54	0.0017	0.0312
p_{stack}	0.0553	7.92	0.0014	0.0312
base random forest	0.0424	11.80	0.0003	0.0312
calibrated random forest	0.0480	10.40	0.0005	0.0312
base extra trees	0.0484	8.77	0.0009	0.0312
calibrated extra trees	0.0448	4.44	0.0114	0.0312
base XGBoost	0.0462	7.04	0.0021	0.0312
calibrated XGBoost	0.0391	7.21	0.0020	0.0312
base KNN	0.1344	8.20	0.0012	0.0312
calibrated KNN	0.1437	10.07	0.0005	0.0312

8.3 Locked test-set results

The locked-test split reverses the cross-validated ranking of the learned fusion system. Although p_{final} performs competitively in cross-validation, it does not achieve the best deployment-side macro F_1. Plain weighted averaging obtains the strongest ensemble macro F_1, while base Extra Trees obtains the best overall combination of macro F_1, Brier score, ECE, and ROC-AUC. This reversal is the central empirical result of the paper: a learned gate trained on out-of-fold probability behaviour can overfit the training-partition meta-distribution and fail under the official UNSW-NB15 train/test shift. The gate still obtains the highest accuracy point estimate and the strongest fusion-path ROC-AUC, but its accuracy advantage over base Extra Trees is small and its bootstrap confidence interval overlaps base Extra Trees in the marginal comparison shown in Table **??**. The macro-F_1 drop between five-fold cross-validation and the locked test set is method-specific: p_{final} drops by 0.143, p_{weighted} drops by 0.072, the tree-based base learners drop by 0.058–0.085, and KNN drops by 0.034–0.036. These results support the interpretation of the gated model as a reliability-oriented diagnostic architecture rather than the preferred deployment classifier for this benchmark. Table **??** summarises the locked-test metrics for all eleven methods.

Table 8 Locked-test multi-class results. The proposed system obtains the highest accuracy point estimate among the evaluated methods (its 95% bootstrap CI overlaps base Extra Trees; see Table **??**) and the strongest ROC-AUC among the three ensemble fusion paths. Plain weighted averaging obtains the highest ensemble macro F_1, while base Extra Trees obtains the best overall Brier score, ECE, ROC-AUC, and macro F_1 among the listed methods. Bold marks the best within the three ensemble fusion paths.

Method	mac. F_1	wgt. F_1	acc.	log loss	Brier	ECE	ROC-AUC
p_{weighted}	**0.5485**	0.7424	0.7678	0.6073	0.0306	0.0649	0.9465
p_{stack}	0.4998	**0.7581**	0.7625	0.7526	0.0336	0.0948	0.9350
p_{final} (**prop.**)	0.5167	0.7547	**0.7685**	0.6114	**0.0306**	0.0695	**0.9475**
base RF	0.5431	0.7420	0.7583	0.8052	0.0313	0.0641	0.9353
cal RF	0.5263	0.7322	0.7608	0.6456	0.0316	0.0784	0.9359
base ExtraTrees	0.5528	0.7553	0.7674	0.6755	0.0294	0.0424	0.9496
cal ExtraTrees	0.5374	0.7396	0.7647	0.6270	0.0307	0.0698	0.9426
base XGBoost	0.5388	0.7364	0.7646	0.6956	0.0324	0.0993	0.9494
cal XGBoost	0.5453	0.7372	0.7647	0.7295	0.0325	0.0971	0.9349
base KNN	0.4887	0.7319	0.7525	4.5124	0.0362	0.1154	0.8199
cal KNN	0.4814	0.7293	0.7528	0.8233	0.0351	0.0761	0.8907

Bootstrap confidence intervals

To avoid over-interpreting small differences between methods on the locked test set, we computed nonparametric percentile bootstrap 95% confidence intervals from $B = 300$ resamples of the test set, fixing the random seed to 42 for exact reproducibility (Table **??**). The accuracy intervals for p_{final} ($[0.7666, 0.7706]$) and base Extra Trees ($[0.7653, 0.7694]$) overlap substantially, so the proposed system's apparent accuracy advantage is not statistically distinguishable from base Extra Trees at the 95% level. Conversely, the gap between p_{stack} macro F_1 ($[0.4958, 0.5038]$) and base Extra Trees macro F_1 ($[0.5453, 0.5596]$) is wide and non-overlapping, confirming the stacker's macro-F_1 disadvantage on the locked split. The gate's improvement over the stacker on macro F_1 (p_{final} $[0.5115, 0.5215]$ vs p_{stack} $[0.4958, 0.5038]$) is also non-overlapping and therefore robust.

8.4 Per-class precision, recall, F_1 and support on the locked test set

Table **??** reports per-class precision, recall, F_1, and support on the locked test set for p_{final}, p_{weighted}, and the strongest single base (Extra Trees). Among the three fusion paths p_{stack} leads on six classes by recall (not shown for space; see Table **??**), the proposed p_{final} leads on `DoS` and ties on `Generic`, and p_{weighted} leads on `Normal` and

Table 9 Nonparametric percentile bootstrap 95% confidence intervals on locked-test metrics ($B = 300$ resamples, seed 42). O
indicate that the corresponding metric difference is not statistically distinguishable at the 95% level on this single test split.

Method	macro F_1	accuracy	Brier	ECE
p_{final}	0.5167 [0.5115, 0.5215]	0.7685 [0.7666, 0.7706]	0.0306 [0.0303, 0.0308]	0.0695 [0.0679, 0.0711]
p_{weighted}	0.5485 [0.5407, 0.5556]	0.7678 [0.7657, 0.7696]	0.0306 [0.0304, 0.0309]	0.0649 [0.0634, 0.0666]
p_{stack}	0.4998 [0.4958, 0.5038]	0.7625 [0.7605, 0.7644]	0.0336 [0.0333, 0.0338]	0.0948 [0.0932, 0.0965]
base ExtraTrees	0.5528 [0.5453, 0.5596]	0.7674 [0.7653, 0.7694]	0.0294 [0.0292, 0.0296]	0.0424 [0.0408, 0.0438]

`Exploits`. Across precision/recall/F_1 jointly, base Extra Trees obtains the highest per-class F_1 on `Worms` (0.595), `Backdoor` (0.137), and `Fuzzers` (0.381); p_{weighted} leads on `Worms` precision (0.659) and `Exploits` F_1 (0.682); and p_{final} leads on `Backdoor` recall (0.135 versus 0.059 for p_{weighted}). The two near-zero-recall classes (`Analysis`, `Backdoor`) are problematic for every method on UNSW-NB15 due to severe overlap with `Exploits`.

Table 10 Per-class precision, recall, F_1, and support on the locked test set for p_{final}, p_{weighted}, and base Extra Trees. Support is the number of test samples in each class.

	p_{final}			p_{weighted}			base Extra Trees			
Class	P	R	F_1	P	R	F_1	P	R	F_1	Support
Normal	0.797	0.976	0.877	0.775	0.985	0.867	0.799	0.972	0.877	56 000
Analysis	0.046	0.002	0.004	0.000	0.000	0.000	0.051	0.015	0.023	2 000
Backdoor	0.111	0.135	0.122	0.981	0.059	0.111	0.126	0.151	0.137	1 746
DoS	0.350	0.747	0.476	0.346	0.643	0.450	0.351	0.741	0.476	12 264
Exploits	0.863	0.541	0.665	0.764	0.617	0.682	0.892	0.519	0.656	33 393
Fuzzers	0.751	0.222	0.343	0.768	0.154	0.257	0.738	0.257	0.381	18 184
Generic	0.977	0.986	0.981	0.961	0.986	0.973	0.962	0.985	0.973	40 000
Reconnaissance	0.893	0.777	0.831	0.926	0.748	0.828	0.866	0.794	0.828	10 491
Shellcode	0.495	0.811	0.615	0.693	0.603	0.645	0.446	0.832	0.581	1 133
Worms	0.150	0.800	0.253	0.659	0.685	0.672	0.479	0.785	0.595	130

8.5 Calibration evidence

XGBoost is by far the best-calibrated raw learner across CV folds (ECE ≈ 0.012 on outer validation), consistent with the dominance of its weight in the reliability-criterion average; KNN is the most miscalibrated raw learner on the locked test set (ECE = 0.115), and OvR isotonic post-correction reduces its locked-test ECE by approximately 0.04 (from 0.115 raw to 0.076 calibrated). Figure **??** reports reliability diagrams for the four base learners on one outer-validation fold (post-OOF, post-calibration in this fold); the diagonal of perfect calibration is overlaid.

Figure **??** extends the calibration analysis to the *locked test set* for the strongest single learner (base Extra Trees) and the three fusion paths. Although the proposed gated system improves fusion-path ROC-AUC and accuracy, base Extra Trees remains better calibrated on the locked test set, indicating that the learned gate is more sensitive to train/test shift than the simpler weighted average.

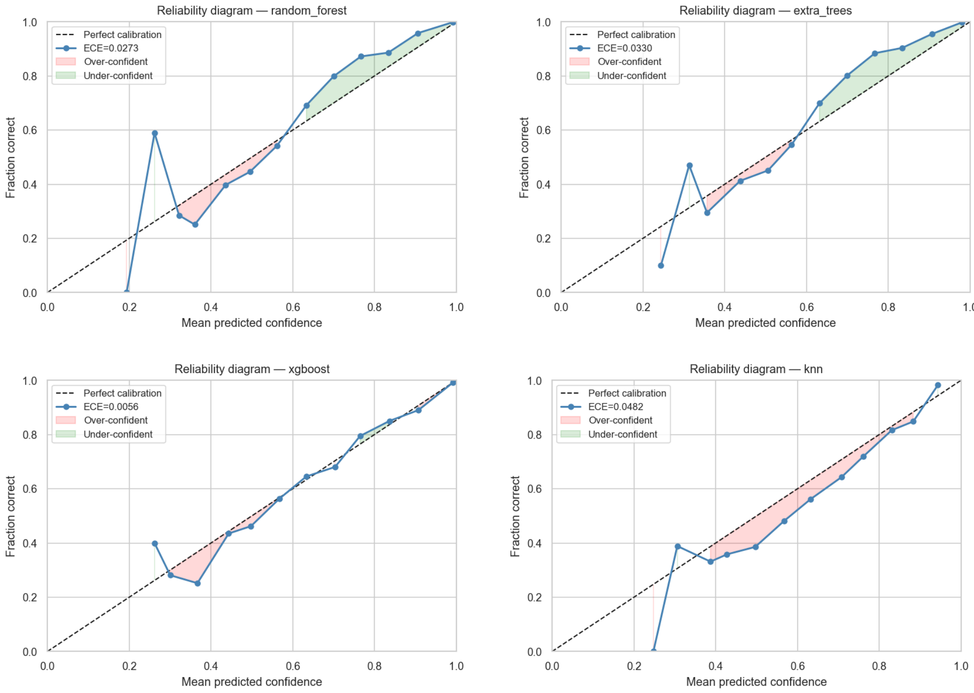

Fig. 2 Reliability diagrams for the four base learners (one outer fold). Each bar represents one max-confidence bin; bar height is the empirical accuracy in that bin. Grey diagonal: perfect calibration.

8.6 Ablation impact

The largest individual contributions to cross-validated macro F_1 come from rich meta-features (+0.045 over the simplified ablation) and the learned gate (+0.038 over weighted-average-only, +0.055 over stacker-only); engineered features and consensus-selector stability weighting contribute smaller but positive effects. Figure **??** visualises the per-configuration result.

8.7 Confusion matrix for the final system

The dominant confusion patterns on the locked test set mirror the known UNSW-NB15 difficulties: `Analysis` and `Backdoor` are largely absorbed into `Exploits`, and `Fuzzers` split between several categories. Figure **??** shows the row-normalised confusion matrix for p_{final}.

8.8 Disagreement-error slices

Of the 175 341 test samples, 95.04% fall in a high-agreement stratum (more than three-quarters of bases agree) and achieve 0.788 accuracy and 0.528 macro F_1; the 4.83% medium-agreement tail drops to 0.412 accuracy and 0.307 macro F_1, and the 0.14% low-agreement tail collapses to 0.169 accuracy and 0.093 macro F_1. This validates the design intuition that most predictions are easy and dominated by base agreement, figures/reliability_test_fusion.png

Fig. 3 Locked-test reliability diagrams (15 max-confidence bins) for the strongest base learner and the three fusion paths. ECE values are reported in the panel titles. Although the proposed gated system improves fusion-path ROC-AUC and accuracy, base Extra Trees remains the best-calibrated method on the locked test set.

while a small high-error tail of disagreement-heavy samples is precisely where the gate has the most leverage. Figure **??** visualises the three strata.

8.9 Security-oriented metrics

ML-style metrics alone are insufficient for IDS evaluation. Table **??** reports balanced accuracy, Matthews correlation coefficient (MCC), binary attack-vs-Normal recall (true positive rate on the 119 341 attack samples), false alarm rate (FAR; predicted-attack rate on the 56 000 Normal samples), and macro PR-AUC, all computed directly from the saved test predictions. The proposed p_{final} has the highest MCC (0.7146) and a lower FAR (0.0240) than p_{stack} (0.0369) and base Extra Trees (0.0279), but plain weighted averaging is the safest in terms of FAR (0.0146) and base Extra Trees

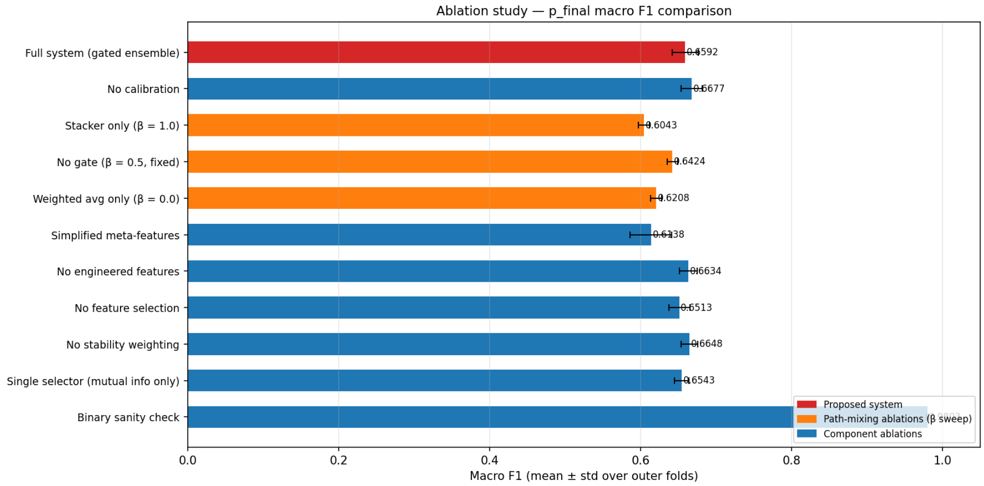

Fig. 4 Cross-validated macro F_1 across the proposed system and nine ablations. Error bars show the standard deviation across the five outer folds.

achieves the highest macro PR-AUC (0.6149). The gate therefore trades a small FAR increase for higher attack recall and MCC relative to the weighted average.

Table 11 Security-oriented locked-test metrics for the three fusion paths and the strongest single base learner. Attack recall and FAR are computed on the binary Normal-vs-Attack collapse of the 10-class predictions ($n_{\text{attack}} = 119\,341$, $n_{\text{normal}} = 56\,000$). All values are computed from the saved locked-test probability arrays.

Method	macro F_1	bal. acc.	MCC	att. recall	FAR	macro PR-AUC	ECE
p_{weighted}	0.5485	0.5479	0.7092	0.8654	0.0146	0.6051	0.0649
p_{stack}	0.4998	0.6189	0.7088	0.8997	0.0369	0.5932	0.0948
p_{final}	0.5167	0.5996	**0.7146**	0.8832	0.0240	0.6005	0.0695
base ExtraTrees	**0.5528**	0.6050	0.7135	0.8854	0.0279	**0.6149**	**0.0424**

8.10 Paired bootstrap differences

Because the locked test set is fixed, paired bootstrap intervals over test samples are more appropriate for comparing methods than visually comparing separate marginal CIs. Table **??** reports paired Δ-CIs ($B = 300$, seed 42) for three pairwise comparisons. The macro-F_1 differences between p_{final} and the two stronger comparators are large and their CIs do not cross zero ($p_{\text{final}} - p_{\text{weighted}} = -0.0318$ with CI $[-0.0375, -0.0263]$; $p_{\text{final}} - \text{ET} = -0.0361$ with CI $[-0.0408, -0.0314]$), confirming that p_{final} has a robust macro-F_1 disadvantage on the locked split. The accuracy advantage of p_{final} over base Extra Trees is small but statistically distinguishable under paired resampling ($\Delta = +0.0011$, CI $[+0.0004, +0.0020]$), which sharpens the marginal-CI overlap noted

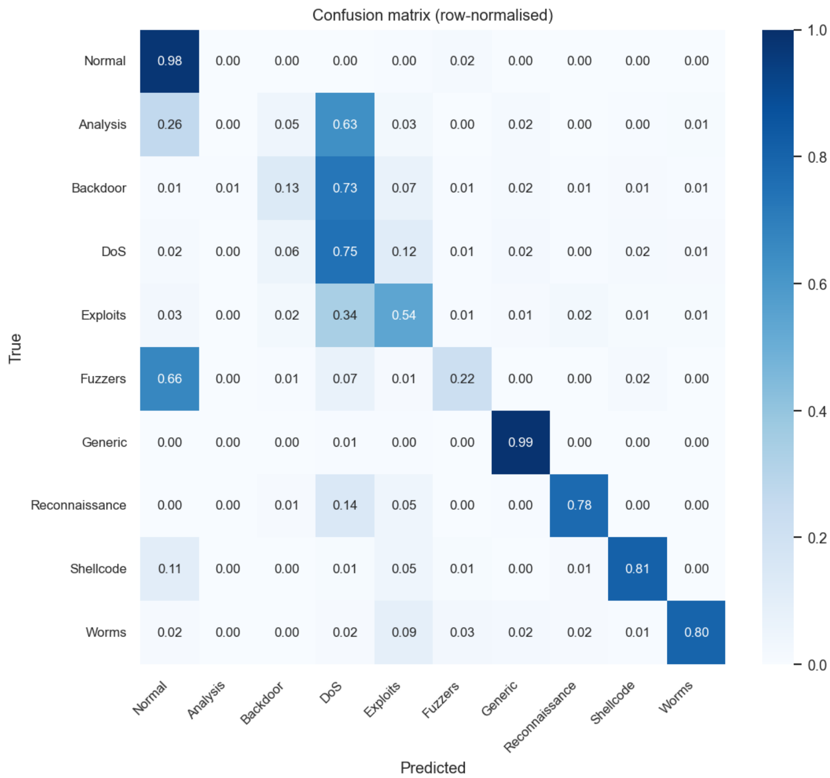

Fig. 5 Locked-test confusion matrix for the proposed system p_{final}. Rows are true classes, columns are predicted; cell intensity is the row-normalised frequency.

earlier. The ECE differences are large and clearly favour base Extra Trees over both fusion paths.

Table 12 Paired nonparametric bootstrap Δ-CIs ($B = 300$, seed 42) for locked-test metric differences. CIs that exclude zero indicate statistically distinguishable differences under paired resampling on the same test instances.

	Δ macro F_1		Δ accuracy		Δ ECE	
Comparison	point	95% CI	point	95% CI	point	95% CI
$p_{\text{final}} - p_{\text{weighted}}$	-0.0318	$[-0.0375, -0.0263]$	$+0.0008$	$[-0.0001, +0.0017]$	$+0.0046$	$[+0.0036, +0.0056]$
$p_{\text{final}} - \text{ET}$	-0.0361	$[-0.0408, -0.0314]$	$+0.0011$	$[+0.0004, +0.0020]$	$+0.0271$	$[+0.0262, +0.0279]$
$p_{\text{weighted}} - \text{ET}$	-0.0042	$[-0.0100, +0.0015]$	$+0.0003$	$[-0.0007, +0.0015]$	$+0.0226$	$[+0.0214, +0.0237]$

8.11 Selective prediction and analyst escalation

The goal of the disagreement-aware pipeline is not necessarily to replace simpler classifiers, but to identify which samples should not be trusted. We evaluate six escalation

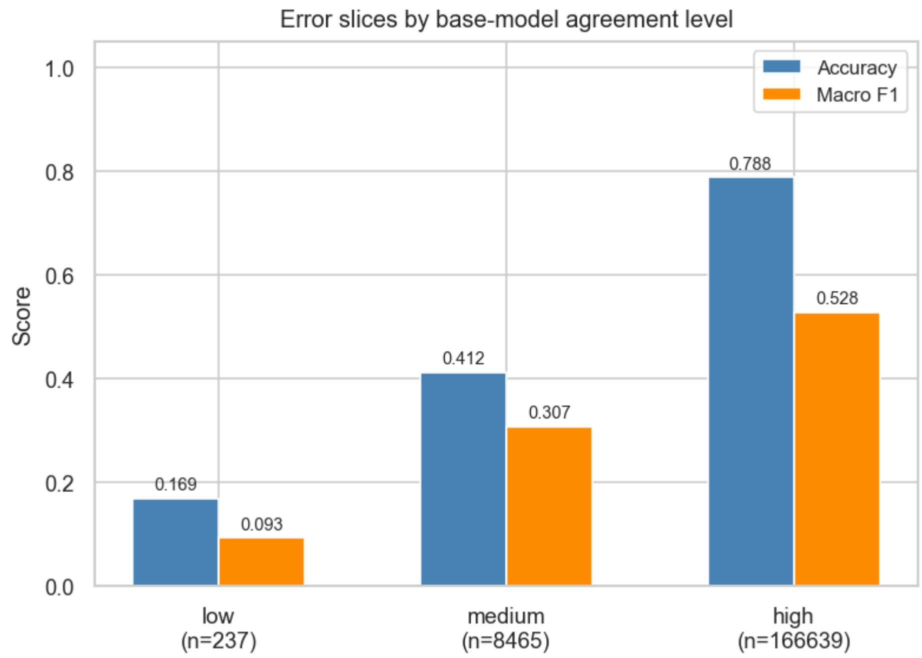

Fig. 6 Locked-test accuracy and macro F_1 of p_{final} stratified by base-learner agreement level.

policies on the locked test set using p_{final} outputs only: predict all; escalate the bottom 5% or 10% of samples by maximum confidence; escalate medium- and low-agreement strata (less than three of four bases agree); escalate the top 5% by Shannon entropy; or escalate the top 5% by absolute path-gap $|\max p_s - \max p_w|$. Table **??** reports retained-sample metrics for each policy, and Figure **??** visualises the trade-off.

Table 13 Selective-prediction policies on the locked test set using p_{final}. Coverage is the retained fraction; metrics are computed on the auto-classified (non-escalated) samples. "Esc. med./low agree" uses the four-base argmax-vote agreement defined in §**??**. All values computed from the saved test predictions.

Policy	Coverage	Accuracy	macro F_1	Att. recall	FAR	Escalated
Predict all	1.000	0.7685	0.5167	0.8832	0.0240	0
Esc. bottom 5% by confidence	0.950	0.7947	0.5424	0.8782	0.0224	8 751
Esc. bottom 10% by confidence	0.900	0.8191	0.5650	0.8720	0.0207	17 532
Esc. med./low agreement	0.950	0.7875	0.5278	0.8762	0.0187	8 702
Esc. top 5% by entropy	0.950	0.7942	0.5435	0.8774	0.0224	8 759
Esc. top 5% by path-gap	0.950	0.7853	0.5289	0.8798	0.0164	8 750

Escalating the bottom 10% of confidence raises retained-sample accuracy from 0.769 to 0.819 and macro F_1 from 0.517 to 0.565, while only modestly reducing binary attack recall on the auto-classified subset (0.883 → 0.872). The path-gap policy yields the lowest FAR (0.0164) at the same coverage. These results show that calibrated figures/selective_prediction.png

Fig. 7 Selective-prediction trade-off: accuracy and macro F_1 on retained (auto-classified) samples for six escalation policies on p_{final}. Coverage labels show the fraction of samples retained.

probabilities and disagreement signals are useful for analyst triage even when the gated point classifier is not the macro-F_1 winner.

8.12 Gate diagnostics on the locked test set

Figure **??** plots the empirical distribution of $\beta(x)$ on the 175 341 locked-test samples. The locked-test mean is 0.467 and the standard deviation is 0.385; 26.7% of samples receive $\beta < 0.01$ (full trust in p_{weighted}), 25.8% receive $\beta > 0.99$ (full trust in p_{stack}), and 32.2% fall in the moderate band $\beta \in (0.4, 0.6)$. The bimodality observed in cross-validation is therefore preserved at deployment time. Stratifying by base-learner agreement (computed in §**??**), the mean β on high-agreement samples ($n = 166\,639$) is 0.469 ± 0.395, on medium-agreement ($n = 8\,465$) is 0.436 ± 0.092, and on low-agreement ($n = 237$) is 0.425 ± 0.029. The gate's variance collapses sharply on disagreement-heavy strata, which suggests the depth-4 tree gate is making homogeneous decisions on small disagreement leaves rather than truly adapting to ambiguity.

figures/beta_diagnostics.png

Fig. 8 Gate $\beta(x)$ diagnostics on the locked test set. *Left*: empirical distribution of β over all 175 341 test samples (mean 0.467, std 0.385); the bimodality observed in cross-validation persists at deployment. *Right*: mean β stratified by base-learner agreement level, with error bars showing the within-stratum standard deviation. The gate's per-sample β variance collapses sharply on the small disagreement-heavy strata.

8.13 Minority-class failure modes and operational risk

The locked-test weakness is concentrated in the rare attack categories rather than the majority classes (Table **??**). The two most damaging failures are *Analysis* ($F_1 \leq 0.023$ for every method evaluated; p_{final} recall 0.002) and *Backdoor* ($F_1 \leq 0.137$; best F_1 is base Extra Trees at 0.137). Both classes are systematically confused with *Exploits*, suggesting that the official UNSW-NB15 split's minority-class samples are feature-overlapping with the dominant attack family. Operationally, this matters: *Analysis* corresponds to suspicious probing or analyst-style probing behaviour, while *Backdoor* corresponds to persistent-access traffic, both of which are cyber-defence priorities. Table **??** summarises the operational implications.

Table 14 Operational risk profile by attack class. The "Best F_1" column is the highest per-class F_1 among the methods reported in Table ?? (the three fusion paths and base Extra Trees) on the locked test set. The rare-class failures concentrate on attack categories with the most direct cyber-defence implications.

Class	Support	Best F_1	Main confusion	Operational risk
Reconnaissance	10 491	0.831	DoS, Exploits	missed pre-attack scanning
Exploits	33 393	0.682	Fuzzers, DoS	under-prioritised active compromise
Fuzzers	18 184	0.381	Exploits	potential pre-exploit fuzzing missed
Shellcode	1 133	0.645	Normal, Recon.	severe false negatives on payload-like behaviour
DoS	12 264	0.476	Exploits	service-disruption misclassified
Worms	130	0.672	Exploits, DoS	rare but high-impact propagation
Backdoor	1 746	0.137	Exploits	missed persistent access
Analysis	2 000	0.023	Exploits	low separability on probing behaviour

9 Discussion

What failed under shift?

On the locked test set p_{final} loses macro F_1 to plain weighted averaging by 0.032 and to base Extra Trees by 0.036 (paired bootstrap CIs in Table ??), and base Extra Trees obtains the lowest Brier (0.0294), the lowest ECE (0.042), and the highest macro PR-AUC. The macro-F_1 drop between cross-validation and the locked split is also the largest for p_{final} (-0.143) versus -0.072 for p_{weighted} and -0.058 to -0.085 for the tree-based base learners.

Why did it fail?

The gate and stacker are trained on out-of-fold meta-feature behaviour from the training partition, so when the disagreement patterns shift between train and test (Tables ?? and ?? document the class- and feature-distribution shift), the learned meta-mapping degrades. The gate's β-distribution remains bimodal on the locked test set (mean 0.467, std 0.385; §??) but its variance collapses sharply on the small disagreement-heavy strata, suggesting the depth-4 tree gate makes homogeneous decisions on small disagreement leaves rather than truly adapting to ambiguity at deployment time.

What should be deployed?

The conclusion is conditional on the operational objective: deploy plain weighted averaging when locked-test macro F_1 is the primary deployment objective; deploy base Extra Trees when calibration, low Brier score, or low ECE dominate; and use the proposed gated pipeline as an escalation layer when the operational priority is to identify uncertain traffic for analyst review. Three ablations (no_calibration, no_stability_weighting, no_engineered_features) obtain marginally higher CV macro F_1 than the full system, but the differences are within fold-level noise (paired t $p \geq 0.073$); they show that reliability-oriented components trade a small amount of hard-label macro F_1 for calibration, auditability, or robustness rather than guaranteed macro-F_1 improvement.

What remains useful?

Calibrated probabilities and disagreement signals support analyst escalation even when the gated point classifier is not the macro-F_1 winner: escalating the bottom 10% of p_{final} confidence raises retained-sample accuracy from 0.769 to 0.819 and macro F_1 from 0.517 to 0.565 (Table **??**), while the path-gap policy yields the lowest false-alarm rate (0.0164).

What does this mean for IDS research?

The principal lesson is that reliability-aware fusion should not be evaluated only by cross-validated macro F_1. Cross-validated fusion gains can fail to translate into deployment-side performance under the official UNSW-NB15 train/test shift, and IDS reviewers should require locked-test or cross-dataset shift evaluation before accepting a fusion improvement as operationally meaningful.

Limitations and threats to validity.

The main threat to validity is that all primary experiments are conducted on UNSW-NB15. The official split is useful because it exposes nontrivial train/test shift, but it cannot establish that the same gating failure or selective-prediction behaviour will appear on CIC-IDS2017, TON-IoT, Bot-IoT, or live enterprise traffic. The results should therefore be read as a controlled reliability case study rather than a universal IDS benchmark claim. We do not evaluate adversarial evasion, data poisoning, or adaptive attackers; an adaptive attacker could attempt to manipulate flow statistics to induce high-confidence misclassification. No payload-level features or malware binary analysis are used; the IDS operates on flow metadata only. No online concept-drift adaptation or live-deployment latency study is performed, calibration is evaluated offline rather than under live drift, and minority classes (Analysis, Backdoor, Shellcode, Worms) make macro F_1 sensitive and per-class estimates unstable.

Reproducibility.

The full configuration files, deterministic seed registry, run manifest, model cards, and fold-level artefacts are organised under artifacts/<run_id>/ and reports/<run_id>/, and a single script reproduces the entire experimental suite end-to-end on the public UNSW-NB15 official splits. An anonymised repository is available to reviewers and contains the experimental code, configuration files, deterministic seed registry, fold indices, selected feature lists, locked-test probability arrays, locked-test hard predictions, and scripts for regenerating every table and figure; it will be de-anonymised upon acceptance. The public UNSW-NB15 files are not redistributed; scripts expect the official train/test CSV files to be placed in the documented data directory.

10 Conclusion and Future Work

This paper presented a leakage-controlled reliability analysis of calibrated, disagreement-aware ensemble fusion for multi-class intrusion detection on UNSW-NB15. The full pipeline combines out-of-fold calibration, reliability-weighted averaging, full-probability stacking, a Brier-optimal sample-level gate, and stability-weighted consensus feature selection. Cross-validation shows that the gated pipeline improves over selected single-model and simple-fusion comparators, but locked-test evaluation gives a more important result: learned gating is more sensitive to train/test distribution shift than simpler fusion. Plain weighted averaging is the stronger fusion choice when macro F_1 is prioritised, while base Extra Trees is the strongest overall deployment candidate in this experiment. The value of the gated pipeline is therefore diagnostic rather than leaderboard-oriented: it exposes how calibration, disagreement, and sample-level mixing behave under shift, and the disagreement and confidence signals it produces meaningfully support analyst escalation, with a 10% confidence-based escalation policy raising retained-sample accuracy from 0.769 to 0.819 and macro F_1 from 0.517 to 0.565. Future work should evaluate cross-dataset generalisation on CIC-IDS2017, TON-IoT and Bot-IoT, adversarial robustness against evasion and poisoning, online recalibration to track concept drift, deployment-time selective prediction integrated with analyst workflows, and substituting deep-learning bases such as CNN-LSTM inside the same gating framework while leaving the calibration and gate components unchanged.

Declarations

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Conflict of interest. The authors declare no conflict of interest.
Data availability. The UNSW-NB15 dataset is publicly available from the Australian Centre for Cyber Security [?].
Code availability. An anonymised repository is available to reviewers and contains the full experimental code, configuration files, deterministic seed registry, fold indices, selected feature lists, locked-test probability arrays, locked-test hard predictions, and scripts for regenerating every table and figure. The repository will be de-anonymised upon acceptance. The public UNSW-NB15 files are not redistributed; scripts expect the official train/test CSV files to be placed in the documented data directory.